

Practice Session 4

Part 1: Sampling Distributions

1.1 Introduction

Recall that a “sample” is a subset of a population. We can use samples to calculate “statistics”, such as the sample mean, median, and standard deviation. If we were to **repeatedly** sample from the same population and calculate the **mean** of each sample, we would obtain what is known as the **sampling distribution** of the **sample mean**. If we were to instead calculate a **proportion** from each sample, we would obtain the sampling distribution of the **sample proportion**.

In general, a sampling distribution is the distribution of a **statistic** across many samples. This is in contrast to the population distribution, which is the distribution of the values of all individuals (or sampling units) in the population.

1.2 Sampling Distribution Terminology

Match each term with its definition.

Terms:

- (a) s
- (b) σ
- (c) Standard Error (SE)
- (d) x
- (e) $\bar{x}$
- (f) μ

Definitions:

- Standard deviation of a population
- Individual unit of the population
- Mean of a population
- Standard deviation of a sample
- Sample mean
- Standard deviation of a sampling distribution

1.3 Generating Sampling Distributions

Let's generate a sampling distribution using polling data on the president's approval rating. Make sure to load the `library(SDS1000)` to have access to the functions for this exercise.

1. Use the `rapprovals()` function to obtain a sample of `n = 1000` individual approval ratings. Use the `get_proportion()` function to obtain the proportion of voters in this sample that approve of the president.

```
# Load the SDS1000 package
library(SDS1000)

# Setting the seed so that we all get the same "random sample"
set.seed(1000)

# your code here #
```

2. Next, use the `do_it()` function from class to generate 10,000 samples, and repeat the above steps to obtain the proportion for each sample. Assign this vector of proportions to the variable `polling_distribution`, and create a histogram of it. Make sure to have an appropriate title.

Challenge: What do values on the x-axis of this histogram correspond to? How does this differ from histograms that we've previously encountered?

```
# your code here #
```

3. Finally, calculate the mean and standard deviation of your collection of sample proportions. What is another name for the standard deviation?

```
# your code here #
```

1.4 Properties of Sampling Distributions

Answer the following True/False questions relating to sampling distributions.

- (a) True/False: As the sample size n decreases, the standard error of a sampling distribution also decreases.
- (b) True/False: The mean of a sampling distribution is always **greater than** the mean of the population.
- (c) True/False: For sample sizes larger than 1, the standard deviation of the sampling distribution of the sample mean is **smaller than** the standard deviation of the population.
- (d) True/False: Assuming that you have an unbiased sampling procedure with a sufficient sample size, the sampling distribution of the sample mean is approximately normally distributed.
- (e) True/False: It is generally better to have a larger standard error than a smaller one.

Part 2: More Practice on Sampling Distributions

2.1 Practice: Baseball Salaries

For this section, we will use the `BaseballSalaries2015` data set from the `Lock5Data` library. Salaries are recorded in millions of dollars.

1. Using the `sample()` function, obtain a sample of $n = 50$ baseball player salaries. See the `help` page for `sample()` for details.

```
library(SDS1000)
library(Lock5Data)
set.seed(1000)

# your code here #
```

2. Now, using the `do_it()` function, repeat this process 10,000 times, and calculate the mean each time. Compare the histogram of these sample means to a histogram of the individual salaries in the population.

```
# your code here #
```

Notice that the individual salaries are strongly right-skewed, yet the sampling distribution of the mean is roughly symmetric and bell-shaped. The population does not have to be normal for the sampling distribution of the sample mean to be approximately normal. Notice also that the x-axes measure different things: the left histogram shows individual players' salaries, while the right histogram shows the means of samples of 50 players (see the Challenge question in 1.3).

3. Using your original sample of 50 salaries from question 1.), calculate a point estimate of the mean salary.

```
# your code here #
```

4. Calculate the standard deviation of your sampling distribution from question 2.) to find the standard error.

```
# your code here #
```

Part 3: Confidence Intervals

3.1 Confidence Interval Terminology

Match each term with its definition.

Terms:

- (a) Point Estimate
- (b) Interval Estimate
- (c) Confidence Interval
- (d) Confidence Level

Definitions:

- A particular kind of interval estimate, produced by a method such that, over repeated samples, a specified percentage of the intervals produced (the confidence level) will contain the population parameter.

- The percentage of confidence intervals that will contain the population parameter of interest over repeated sampling. For example, a 95% confidence level means that, in the long run, 95% of the confidence intervals generated by the method will contain the population parameter of interest.
- A single sample statistic that we use to estimate a population parameter. The sample mean is an example, as we use it to estimate the population mean.
- A range of plausible values for a population parameter, rather than a single number. It does not by itself carry a stated confidence level. Many interval estimates take the form point estimate $\pm$ margin of error, in which case the point estimate sits at the center.

3.2 Confidence Interval for Baseball Salaries

For this section, we will use the `BaseballSalaries2015` data set from the `Lock5Data` library.

1.) Using the `sample()` function, obtain a sample of $n = 100$ baseball player salaries. Make sure to sample **without replacement**. See the `help` page for `sample()` for details.

```
library(Lock5Data)
set.seed(1000)

# your code here #
```

2.) Now, using the `do_it()` function, repeat this process $n = 10,000$ times, and calculate the mean each time.

```
# your code here #
```

3.) Using your original sample of 100 salaries from part 1.), calculate a point estimate of the mean salary.

```
# your code here #
```

4.) Calculate the standard deviation of your sampling distribution from part 2.) to estimate the standard error.

```
# your code here #
```

5.) Using your answers from part 3.) and 4.), generate a 95% interval estimate for the mean salary of baseball players. The interval estimate is of the form:

Point Estimate $\pm$ 1.96 $\cdot$ Standard Error

.

```
# your code here #
```

6.) Interpret your interval estimate.

7.) Compare your interval estimates to the actual mean salary from the entire data set. Comment on what you notice.

```
# your code here #
```